

Practice Questions

31 July 2026 06:04

Question 1

(i) Absorption law states that:

[1]

(a) $A \cdot (A' + B) = A$ ~~X~~

☒ (b) $A + (A \cdot B) = A$ $\rightarrow$

$0 + AB$
 $A(1+B) = A$

(c) Both (a) and (b)

(d) $A \cdot (B + C) = A \cdot B + A \cdot C$

(ii) **Assertion** : $A=0$ $B=1$ $C=0$ and $D=1$ and minterm is $A' \cdot B \cdot C' \cdot D$

[1]

Reason : The final sum term must be 0 so A and C are complemented.

$\rightarrow \bar{A} + B + \bar{C} + D =$

Which one of the following options is correct?

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

☒ (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

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(iii) According to the Principle of duality, the Boolean equation $(P + Q') \cdot R \cdot 1 = P \cdot R + Q' \cdot R$ will be equivalent to:

[1]

(a) $P \cdot Q' + R + 1 = (P + R) \cdot (Q' + R)$

☒ (b) $P \cdot Q' + R + 0 = (P + R) \cdot (Q' + R)$

(c) $P' \cdot Q + R + 1 = (P' \cdot R') \cdot (Q + R')$

(d) $P \cdot Q' + R \cdot 0 = (P + R) \cdot (Q' + R)$

(iv) The complement of the Boolean expression $(X \cdot Y)' + Z'$ is:

[1]

(a) $(X + Y) \cdot Z$

☒ (b) $X \cdot Y \cdot Z$

(c) $(X' + Y') \cdot Z'$

(d) $(X' + Y') \cdot Z$

(v) The equivalent of $(P \wedge Q) \vee (\sim P \wedge \sim Q)$ will be:

[1]

(a) $((P \wedge Q) \vee \sim P) \wedge \sim Q$

☒ (b) $(P \wedge Q) \vee (\sim P \wedge \sim Q)$

(c) $P \wedge (Q \vee \sim P) \wedge \sim Q$

(d) $P \wedge (Q \vee (\sim P \wedge \sim Q))$

(vi) **Assertion** : Boolean algebra and Binary number system are different from each other.

[1]

Reason : There are some basic operations like AND, OR and NOT which are performed only in Boolean algebra.

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

performed only in Boolean algebra.

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
- ✓ (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

Question 3

- (i) A Football Association coach analyses the criteria for a win/draw of his team [5] depending on the following conditions.

- If the Centre and Forward players perform well but Defenders do not perform well.

OR

- If Goalkeeper and Defenders perform well but the Centre players do not perform well.

OR

- If all perform well.

The inputs are:

INPUTS	
C	Centre players perform well
D	Defenders perform well
F	Forward players perform well
G	Goalkeeper perform well

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: **X** - Denotes the win/draw criteria [1 indicates win/draw and 0 indicates defeat in all cases]

Draw the truth table for the inputs and outputs given above and write the SOP expression for X(C, D, F, G). *output: '1' is selected here.*

C	D	F	G	X	
0	0	0	0	0	
0	0	0	1	0	
0	0	1	0	0	
0	0	1	1	0	
0	1	0	0	0	
0	1	0	1	1	✓
0	1	1	0	0	
0	1	1	1	1	✓
1	0	0	0	0	
1	0	0	1	0	
1	0	1	0	1	✓
1	0	1	1	1	✓
1	1	0	0	0	
1	1	0	1	0	
1	1	1	0	0	
1	1	1	1	1	✓

$$\rightarrow \text{SOP } X(C, D, F, G) = C'DF'G + C'DFG + C'D'F'G + C'D'FG + CDFG$$